



**Kampus
Merdeka**
INDONESIA JAYA

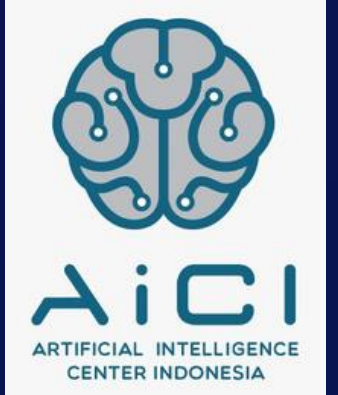
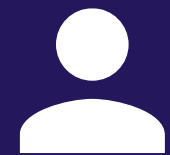


Image Basic with OpenCV



Penyusun Modul : Febri Faresi

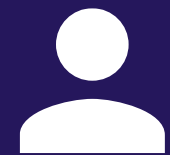
Editor : Citra Chairunnisa



OpenCV

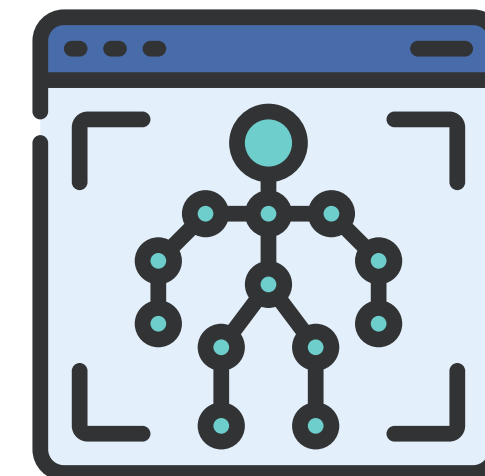
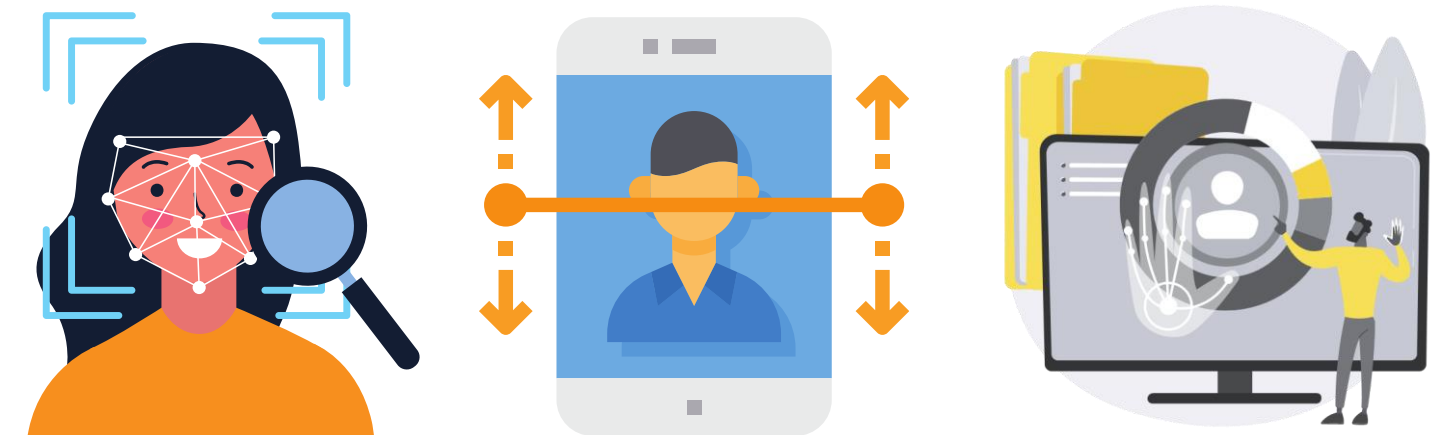
- OpenCV (Open-Source Computer Vision Library) is a library of programming functions mainly aimed at real-time computer vision.
- Originally developed by Intel in 1999, it was later supported by Willow Garage then Itseez (which was later acquired by Intel).
- It is written in C++ (we will be using the Python bindings).
- The library is cross-platform and free for use under the open-source Apache 2 License.

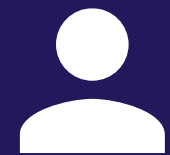




Various applications...

- Human-Computer Interaction (HCI)
- Object Identification
- Object Recognition
- Face Recognition
- Gesture Recognition
- Motion Tracking
- Image Processing
- Mobile Robotics





Why OpenCV?

Best competitor: **Matlab**



Cons

- OpenCV does not provide the same ease of use when compared to Matlab
- Integrated development environment
- Memory management

Pros

Specific

- OpenCV was made for image processing
- Matlab is quite generic

Speed

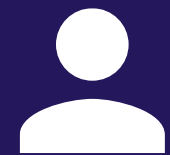
- Around 30 fps in real time image processing (OpenCV)
- Around 4-5 fps in real time image processing (Matlab)

Efficient

- Matlab needs more system resources than OpenCV

Free

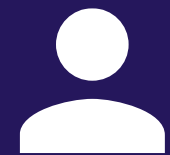




Modules

OpenCV has a modular structure, i.e., the package includes several shared or static libraries:

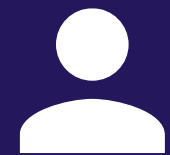
- **core** – basic structure & algorithms
- **imgproc** – image processing algorithms
- **video** – video analysis
- **highgui** – built-in simple UI
- **calib3d** – camera calibration & 3D reconstruction
- **features2d** – 2D features framework
- **objdetect** – detection of obj and other items
- **ml** – ML classes used for classification, regression and clustering data
- **gpu** – GPU-accelerated algorithms



Digital Image

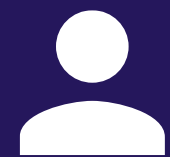
Digital images are electronic photos taken of a scene or scanned from documents. These images are composed of pixels and each pixel is assigned a tonal value (black, white, shades of gray, or color).

- Bit depth
- Black and white images
- Color images
- Binary or bilevel images
- Resolution
- Pixel dimensions
- File size



Bit Depth

Bit depth is determined by the number of bits used to define each pixel. The greater the bit depth, the larger the number of tones (grayscale or color) that can be represented. 1 bit can represent two tones, typically black and white. A 2-bit image has four possible combinations: 00, 01, 10, 11, which represent black, dark gray, light gray, and white. The number of tones that can be represented for a 2-bit image is $2^2 = 4$. In other words, an 8-bit image has 2^8 different tones that can be assigned to each pixel.

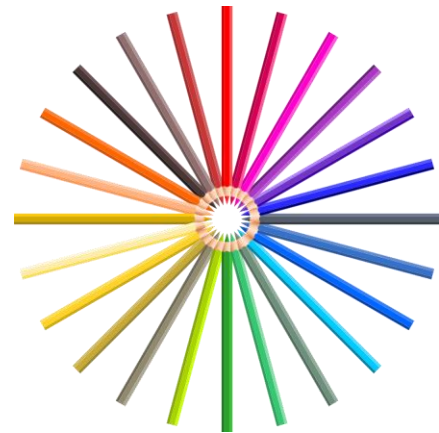
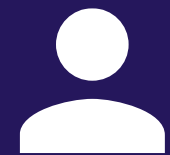


Black and White Images

Black and white images are made up of pixels each of which hold gray levels which span the full range from black to white. Assuming 256 gray levels, each black and white pixel can be stored in a single byte (8 bits) of memory.

$$2^8 = 256$$

GRAY = 1 SET OF DIGITS		
11111111	11100110	11001101
10110100	10011011	01110011
01010000	00101000	00000000



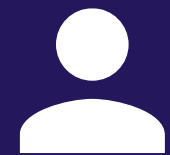
Color Images

Color images are made up of pixels each of which holds three numbers corresponding to the red, green, and blue (RGB) levels of the image at a particular location. Assuming 256 levels for each RGB primary level, each pixel can be stored in three bytes (24 bits) of memory. This corresponds to roughly 16.7 million different possible colors.

$$2^8 \text{ of Red} \times 2^8 \text{ of Green} \times 2^8 \text{ of Blue} = 2^{24} = 16,777,216$$

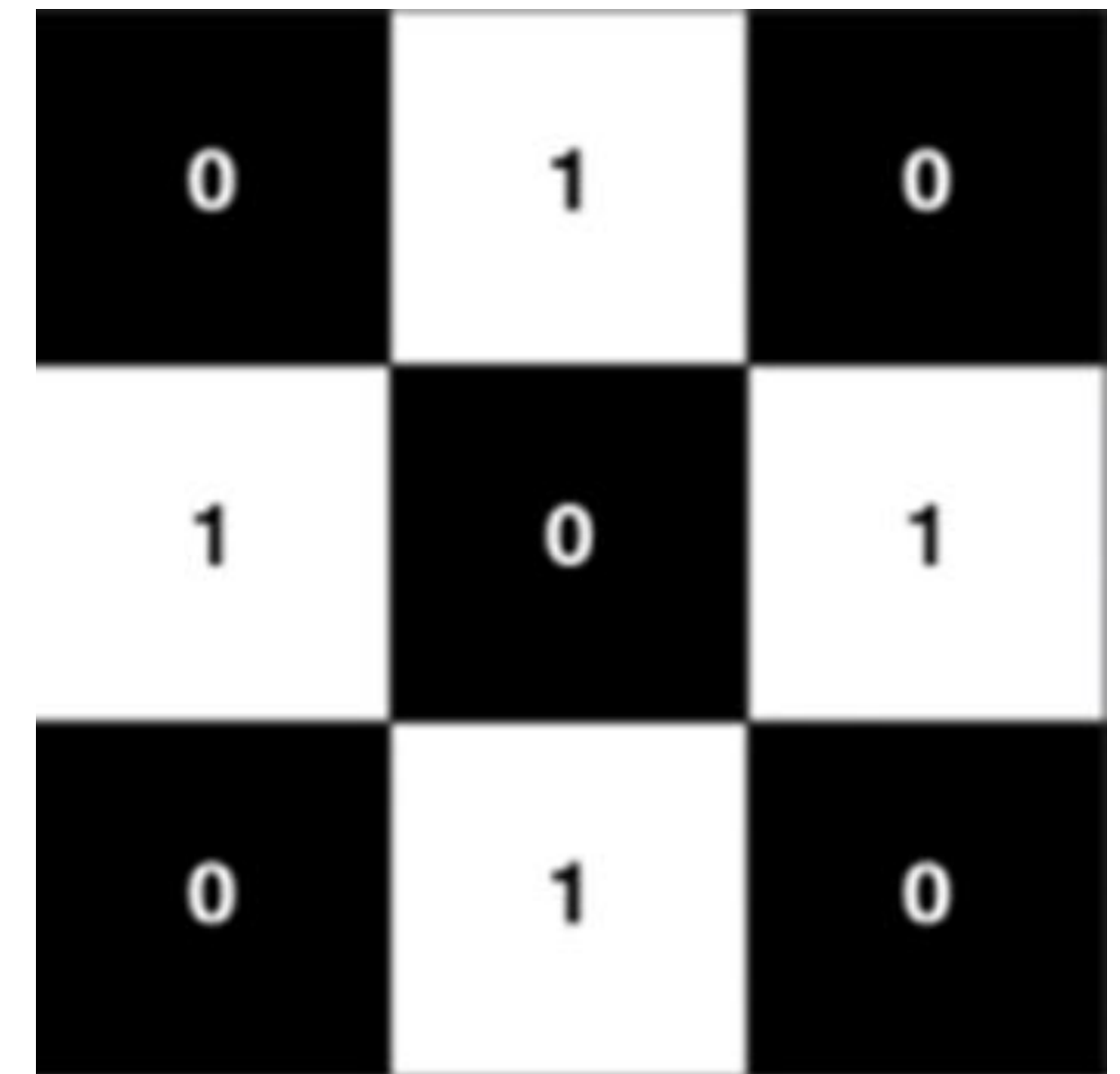
Note that for images of the same pixel size, a black and white version will use three times less memory than a color version.

'RGB' = 3 SETS OF DIGITS		
11111111	01100110	00110011
00000000	01100110	11001100
00000000	11111111	10011001
11111111	11111111	00110011
11111111	00000000	11001100
01100110	11001100	11111111
00110011	00110011	11111111
00110011	00110011	10011001
00000000	10011001	10011001

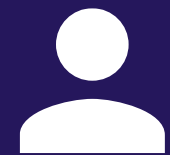


Binary or Bilevel Images

Binary images use only a single bit to represent each pixel. Since a bit can only exist in two states—on or off, every pixel in a binary image must be one of two colors, usually black or white. This inability to represent intermediate shades of gray limits their usefulness in dealing with photographic images.

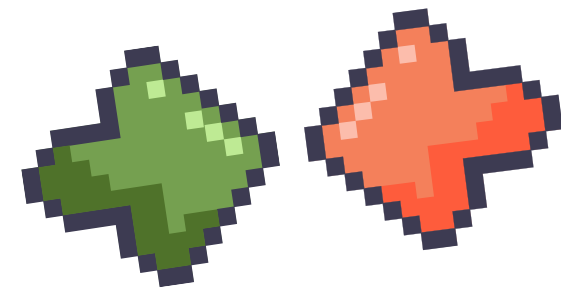
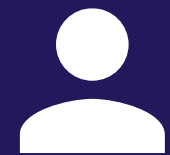


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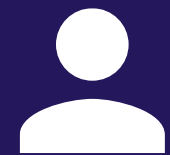
Resolution

Resolution is the ability to distinguish fine detail. The higher the resolution, the more information the image contains. We use dpi (dots per inch) to refer to the quality of an image. A 300dpi image could be a good quality image in that you can see fine detail. However, pixels have no set size as they expand or contract to fill the space available.



Pixel dimensions

Pixel dimensions are the horizontal and vertical measurements of an image expressed in pixels. For example, an 8" x 10" photo that is scanned at 300dpi has the pixel dimension of 2400 x 3000
(8 x 300 = 2400, 10 x 300 = 3000).



File Size

File size is determined by the number of pixels so it is important to know how many pixels the image contains. An image with higher resolution will have a bigger file size. A color image will also have a bigger file size than a grayscale image.

1 Byte



8 bits

1 Kilobyte (KB)



1024 bytes

1 Megabyte (MB)



1024KB

1 Gigabyte (GB)



1024 MB